Predicting Alzheimer's Disease Risk Through Machine Learning: Analyzing Cardiovascular and Metabolic Factors

David Osoro
The Anuradha and Vikas Sinha Department of Data Science
University of North Texas
Denton, TX, USA
david.osoro@my.unt.edu

Sweta Gautam
The Anuradha and Vikas Sinha Department of Data Science
University of North Texas
Denton, TX, USA
sweta.gautam@my.unt.edu

Anish Sigdel
The Anuradha and Vikas Sinha Department of Data Science
University of North Texas
Denton, TX, USA
anish.sigdel@my.unt.edu

Dr. Stephen F. Wheeler
The Anuradha and Vikas Sinha Department of Data Science
University of North Texas
Denton, TX, USA
stephen.wheeler@unt.edu

***Abstract*—Alzheimer's disease (AD) is a progressive neurological disorder that affects millions of people worldwide and is becoming increasingly common as populations age. Early detection is crucial for improving patient outcomes, yet traditional research has often relied on genetic scores or limited statistical models that overlook the complex, non-linear interactions of modifiable health factors.**

This study proposes a machine learning approach to predict AD risk using cardiovascular and metabolic indicators. The input features for this study included cardiovascular measures such as SystolicBP, DiastolicBP, CholesterolTotal, CholesterolLDL, CholesterolHDL, CholesterolTriglycerides, Hypertension, and CardiovascularDisease. Additionally, metabolic measures included BMI, Diabetes, DietQuality, PhysicalActivity, and SleepQuality.

This study utilized a real-world dataset of 2,149 patients, and trained and evaluated multiple models including XGBoost, Random Forest (with Optuna hyperparameter tuning), and a multilayer perceptron deep learning model. A stacking ensemble model was also implemented to combine predictive strengths across individual classifiers. The ensemble model achieved the best performance with an accuracy of 94.9%, recall of 90.8%, and F1 score of 0.931, demonstrating superior ability to identify at-risk individuals.

Feature importance scores were used to compute personalized risk profiles across cardiovascular and metabolic domains, with the final output being a personalized risk score for each patient. The results affirm that machine learning can meaningfully enhance Alzheimer's risk prediction by capturing complex patterns in modifiable health data, offering more inclusive, interpretable, and actionable tools for early intervention and public health planning.

***Index Terms*—*Index Terms*—Alzheimer's Disease (AD): neurodegenerative disorder; Amyloid Plaques: abnormal protein buildup; Neurofibrillary Tangles: tau protein accumulations; MMSE: cognitive screening tool; ADLs: basic daily functions; Chronic Inflammation: sustained immune response; Oxidative Stress: cellular damage from free radicals; Domain-Based Risk Score: model-based risk estimation using health data.**

I. Introduction

Alzheimer's disease (AD) is a progressive neurodegenerative disorder that affects memory, thinking, and behavior. Over time, it causes brain cells to deteriorate and die, leading to a gradual decline in cognitive abilities and, eventually, death among older adults (Gadhave et al., 2024). By 2030, approximately 1 in 5 adults over the age of 65 in the United States is projected to be affected by the disease (Abeysinghe et al., 2025). Despite the rapid increase in Alzheimer's cases, the development and progression of the disease remain poorly understood, and current diagnostic methods are often limited to genetic assessments or rely on broad statistical associations.

This study aims to address these gaps by focusing on health indicators that can be changed through medical or lifestyle interventions. It specifically examines cardiovascular and metabolic markers as predictors of Alzheimer's disease risk. These factors are commonly assessed in clinical practice and represent valuable opportunities for early intervention and prevention. Using a real-world dataset of 2,149 patients from Kaggle.com, this study aims to develop predictive models capable of identifying complex relationships among these variables.

To accomplish this, the project applies several machine learning algorithms, including multivariable regression, Random Forest optimized with Optuna, XGBoost, and a multilayer perceptron deep learning model. A stacking ensemble model is also implemented to combine the strengths of the individual models. These techniques are well suited for detecting non-linear interactions and for ranking the importance of features that contribute to Alzheimer's risk, a capability that traditional models often lack (Liang et al., 2025).

The primary output of the study is a personalized risk

score for each patient. This score is computed using the weighted contribution of both cardiovascular and metabolic features. The final scores are grouped into four risk levels: low, moderate, high, and severe. This structure helps simplify interpretation and increases usefulness in clinical settings. By providing individual predictions, the model supports more focused prevention strategies and allows both healthcare providers and patients to take early action.

Overall, this project presents a meaningful opportunity to apply machine learning to a real-world healthcare challenge. In contrast to prior studies that primarily rely on logistic regression or other linear approaches (Modhugu & Ponnusamy, 2024), our work seeks to model the complex, non-linear dynamics of Alzheimer's progression using advanced techniques. By generating personalized and interpretable risk scores, the study offers practical utility for both clinicians and patients. Ultimately, we aim to contribute to a deeper understanding of how cardiovascular and metabolic health factors influence the risk of developing Alzheimer's disease.

II. Problem Statement

Between 1990 and 2021, global death rates from Alzheimer's disease and related dementias nearly doubled, rising to 15.26 per 100,000 for men, 31.56 for women, and 23.54 overall. Cardiovascular disease has significantly contributed to this increase, accounting for an additional 1.84 per 100,000 for men, 3.94 for women, and 4.70 overall (Mobaderi et al., 2024). Beyond its rising mortality rate, Alzheimer's disease places a growing emotional and financial burden on families, caregivers, and healthcare systems. In 2016, the average cost of formal care for an Alzheimer's patient in the United States was $28,078, while informal care ranged from $15,792 to $36,667 per year (Nandi et al., 2024). These challenges are expected to intensify as the aging population continues to grow.

Given the scale of the issue, there is an urgent need for more effective strategies to detect and manage Alzheimer's disease early. While previous research has shown links between metabolic and cardiovascular factors and Alzheimer's risk, many existing studies have key limitations. These include an overreliance on genetic risk scores, use of linear statistical models that overlook complex interactions, limited sample diversity, and a lack of personalized outputs that can support clinical decision-making. Most studies do not fully explore how multiple modifiable health factors interact to influence risk (L. Liu et al., 2024).

This study aims to address these gaps by using detailed clinical records that include cardiovascular and metabolic indicators. The study applies advanced machine learning techniques, including Random Forest and XGBoost, to model nonlinear relationships between these features and Alzheimer's risk. These models are well suited for identifying complex patterns and determining the relative importance of each feature (FIND A PAPER).

The ultimate goal is to generate personalized Alzheimer's risk assessments that are interpretable and actionable. These insights can guide clinicians and patients in making informed decisions about prevention and care. By improving early identification of high-risk individuals, this research has the potential to advance public health strategies, support clinical tools, and contribute meaningfully to the growing field of data-driven healthcare. The central hypothesis of this study is that individuals with elevated cardiovascular and metabolic risk profiles have a significantly higher likelihood of developing Alzheimer's disease. Addressing this problem has the potential to improve clinical decision-making, empower patients to take preventive steps, and reduce the overall burden of the disease on society.

III. Related Works

Given the rising global impact of Alzheimer's disease, machine learning has become a key tool for modeling and understanding its progression. The article, *Association of Polygenic Risk Score for 5 Diseases with Alzheimer's Disease Progression*, examined whether polygenic risk scores for five common diseases were linked to Alzheimer's biomarkers (Liang et al., 2025). Polygenic risk scores combine many small genetic variations to estimate a person's genetic risk for a disease. The study found that individuals with higher polygenic risk scores for coronary artery disease, type 2 diabetes, and obesity had stronger associations with Alzheimer's biomarkers. This suggests that genes tied to cardiovascular and metabolic health conditions may influence the progression of Alzheimer's disease(Liang et al., 2025). These findings laid the foundation for our research, which focuses on how cardiovascular and metabolic health factors may contribute to the risk and progression of Alzheimer's disease. Understanding these connections could offer valuable insights into early risk detection and personalized intervention strategies for Alzheimer's disease.

While the study offered valuable insights, it also had important limitations that restricted the depth of its findings. One notable limitation was the reliance on simple machine learning models like logistic regression. Logistic regression assumes a non-linear relationship between the features and the outcome. This assumption makes it less effective when the data has complex or non-linear patterns, which are common in health datasets. Additionally, logistic regression is sensitive to how features are scaled. It can also overfit the data when there are too many features. Lastly, logistic regression may struggle to achieve high recall rates, meaning it can miss many true positive cases (Modhugu & Ponnusamy, 2024). To overcome these challenges, our study uses more advanced machine learning techniques. These include XGBoost, Random Forest, Multilayer Perceptron, and Ensemble Learning. The XGBoost model is oftentimes used to model complex relationships between features. The model is also known to help prevent overfitting and it efficiently handles missing or imbalanced data. (Jarupunphol et al., 2025). Additionally, the Random Forest model is known for being robust, reducing overfitting, and handling high-dimensional datasets well. Random Forest models often achieve high accuracy, precision, and F1 scores in healthcare applications (Jarupunphol et al., 2025).

Overall, the implementation of complex models in our study is expected to produce more detailed, reliable, and clinically meaningful results.

Another limitation of the study is that it does not provide patients with personalized risk estimates, which significantly reduces its practical value in clinical settings. Without individualized predictions, it is difficult for patients and healthcare providers to fully understand a person's specific risk of developing Alzheimer's disease. Personalized results are essential because they enable clinicians to tailor prevention strategies and interventions to each patient's unique health profile. They also empower patients to make more informed decisions regarding lifestyle changes, treatment options, and long-term care planning. Our study addresses this gap by generating personalized risk assessments, making the findings more actionable and applicable to real-world healthcare.

Another limitation of the previous study is that it focused exclusively on non-Hispanic White participants. Genetic risk factors and their effects can vary across different populations. Excluding diverse groups may overlook important patterns. In contrast, our study includes participants from a broader range of ethnic backgrounds, including Caucasian, African American, Asian, and others. By incorporating greater diversity, we aim to improve the accuracy, inclusiveness, and real-world applicability of our results. This approach helps ensure that the risk assessments we generate are meaningful for a wider and more representative population.

Another Alzheimer's study, titled *Metabolic Dysfunctions Predict the Development of Alzheimer's Disease: Statistical and Machine Learning Analysis of EMR Data*, examines how metabolic problems and demographic factors like race, ethnicity, and sex influence the risk of developing Alzheimer's disease (AD). The study used electronic medical records (EMR) from six University of California hospitals. Key features analyzed included body mass index (BMI), liver health markers, and alcohol consumption. The goal of the research was to understand how these metabolic and demographic factors contribute to Alzheimer's risk and to build predictive models for early detection of high-risk individuals. Researchers applied both traditional statistical techniques, such as Cox proportional hazards models, and machine learning methods to uncover patterns and predict Alzheimer's risk. The findings showed that Hispanic/Latino and Native Hawaiian/Pacific Islander individuals had a higher risk of developing Alzheimer's compared to White individuals. The study also revealed that underweight individuals were more likely to develop Alzheimer's than those who were overweight or obese. Furthermore, alcohol abuse and non-infectious hepatitis emerged as important risk factors, with alcohol abuse having a stronger impact among women. Overall, the study highlights the critical role of metabolic health in predicting Alzheimer's disease. It also suggests that prevention strategies should be tailored to individuals' metabolic profiles and demographic backgrounds to better address differences across diverse populations (R. Liu et al., 2024).

One limitation of the study is its reliance on diagnostic codes from electronic medical records (EMRs). EMRs are digital records maintained by healthcare providers and are valuable for improving data access, streamlining administrative tasks, and supporting patient care (Honavar, 2020). However, they only reflect conditions that have been formally diagnosed and recorded during medical visits. For example, conditions like alcohol abuse are only documented if a clinician makes an official diagnosis. In reality, many individuals with harmful drinking behaviors never seek care or receive a formal diagnosis, leading to underreporting in EMR-based studies. This under reporting can result in incomplete data and reduce the accuracy of models built solely on diagnostic codes.

To address this limitation, our study uses patient datasets that include detailed clinical measurements, such as cholesterol levels, blood pressure, and body mass index (BMI). These objective, continuous health indicators provide a more comprehensive view of a patient's cardiovascular and metabolic status. By incorporating this richer data, our machine learning models can make more accurate risk predictions and offer a more reliable assessment of Alzheimer's disease risk.

The study also encountered challenges related to sample selection and population representation. It included only patients who had received care at University of California hospitals before the age of 69 and who remained alive without an Alzheimer's diagnosis by age 70. This approach likely excluded individuals who developed Alzheimer's at a younger age or those who had limited access to healthcare services. Additionally, while the study included participants from various racial and ethnic backgrounds, minority groups were still underrepresented, limiting the generalizability of the findings. In contrast, our study aims to improve representation by including participants from a broader range of ethnic groups, such as African American, Asian, and Caucasian populations. Furthermore, instead of relying only on group-level averages, we will generate personalized risk assessments. This personalized approach allows for more precise, actionable insights that can better support individualized prevention and care strategies for diverse patient populations.

IV. Objectives of the Study

The primary objective of this study is to investigate how cardiovascular and metabolic health indicators influence the risk of developing Alzheimer's disease. Much of the existing research on Alzheimer's disease has focused on genetic factors or neurological conditions. However, recent studies show that cardiovascular and metabolic issues, such as blood pressure problems, cholesterol imbalances, diabetes, and body weight changes, also play an important role. Unlike genetic traits, these cardiovascular and metabolic factors can often be managed or improved over time through lifestyle changes, medication, or other medical interventions. This study aims to highlight how these everyday health markers are related to Alzheimer's risk. The goal is to support early detection and help create personalized strategies that may lower the chance of developing the disease. To achieve this, the study will apply advanced machine learning and deep learning

models to uncover complex patterns in health data. It will also compare the efficiency, accuracy, and interpretability of multiple models to determine the best-performing approach. Finally, the study will generate personalized Alzheimer's risk scores that account for each individual's cardiovascular and metabolic health indicators.

To achieve its goals, this study applies advanced machine learning and deep learning techniques to uncover patterns in the data that may not be immediately apparent. The models used include Random Forest, XGBoost, and a Multilayer Perceptron (MLP), each chosen for its ability to handle complex, nonlinear relationships. Additionally, a stacking ensemble model is implemented to combine the strengths of these individual models and improve overall predictive performance. Complex models like these are especially useful when health factors interact in ways that are not simple. For example, a patient with both high blood pressure and high BMI might be at a greater risk for Alzheimer's disease than someone with only one of those conditions. Relationships like these are often missed by traditional statistical models or simple machine learning approaches that treat each feature separately. Studies have shown that machine learning models can uncover nonlinear relationships in health data that are not visible through standard analysis methods (Ibrahim et al., 2020). By using more complex models, this study aims to capture the subtle and combined effects of different health indicators. This can lead to better predictions, improved early detection, and a deeper understanding of how different health profiles relate to Alzheimer's risk.

As part of the study's objectives, each selected machine learning model will be trained and evaluated systematically. Comparing the models will help determine how well they perform on real-world patient data. By analyzing outcomes such as prediction accuracy, consistency across different patient subgroups, and computational efficiency, the study aims to understand the strengths and limitations of each method. This evaluation will not only help identify the best-performing model but also provide insights into which approaches are most suitable for clinical use and public health applications.

The final key objective of this study is to create personalized risk scores for each patient. This is important because every person is different. Instead of using the same prediction for everyone, the model gives a score that reflects each person's unique health data. This makes the results more useful and easier to apply. Doctors can better understand a patient's risk and give advice or treatment that fits their specific needs. Clear communication of personalized risk is also critical. Research shows that when risk information is presented in simple and understandable formats, patients are more likely to interpret the information correctly and take informed action (National Institute for Health and Care Excellence, 2021). By providing clear and personalized risk scores, this study aims to make the results more actionable in real-world healthcare settings and better support shared decision-making between patients and providers.

V. Hypothesis

The *formal hypothesis* guiding this research is that machine learning models trained on cardiovascular and metabolic features can create personalized risk scores that accurately predict a person's likelihood of developing Alzheimer's disease.

The *alternative hypothesis* (H_1) proposes that cardiovascular and metabolic features contain useful information that allows machine learning models to accurately estimate a person's Alzheimer's disease risk.

The *null hypothesis* (H_0) states that cardiovascular and metabolic features do not contain enough predictive information to accurately estimate Alzheimer's disease risk using machine learning models.

To test these hypotheses, statistical methods such as the independent samples t-test and the Mann-Whitney U test will be used to evaluate differences in risk scores between individuals diagnosed with Alzheimer's disease and those without. This evaluation will be presented in the later sections of the project.

VI. Research Design and Methodology

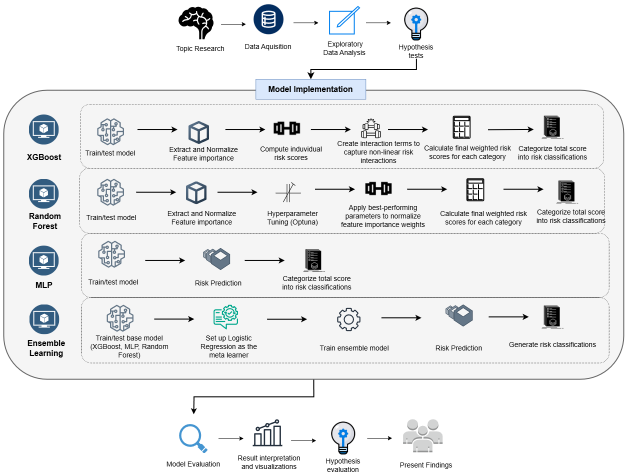

Figure 1. Study design and model implementation framework. The flowchart outlines the research workflow, starting from topic research, data acquisition, and exploratory data analysis, followed by hypothesis testing and model implementation. Models used include XGBoost, Random Forest, and Multilayer Perceptron (MLP). Ensemble learning is applied to combine model predictions. The final steps involve model evaluation, hypothesis evaluation, and result interpretation.

A. Dataset and Feature Selection

The dataset for this study was selected from Kaggle because it included a wide range of real-world patient health records. It provided detailed information about cardiovascular and metabolic health, which are important areas of research for Alzheimer's disease risk. This made it a strong choice for this project, as it allowed us to focus on everyday health measures that can be monitored and changed over time, rather than relying only on genetic or neurological markers.

We carefully reviewed the available features and selected those that have been shown in recent studies to have a meaningful impact on Alzheimer's disease. For cardiovascular features, we used systolic blood pressure, diastolic blood pressure, total cholesterol, LDL cholesterol, HDL cholesterol, triglyceride levels, hypertension, and cardiovascular disease. High blood pressure, particularly during midlife, is associated with damage to the blood-brain barrier and reduced blood flow, both of which are known to increase Alzheimer's disease risk (Pacholko & Iadecola, 2024). Cholesterol also plays a major role, with high LDL cholesterol and high triglyceride levels linked to greater risk of Alzheimer's, while high HDL cholesterol appears to have a protective effect on brain health (Crichton et al., 2014; Sáiz-Vazquez et al., 2020). Cardiovascular disease itself is an important feature because individuals with a history of heart problems have been shown to face a higher risk of developing Alzheimer's disease (Gottesman et al., 2017).

For metabolic features, we selected body mass index (BMI), diabetes status, diet quality, physical activity, and sleep quality. Diabetes has been strongly linked to Alzheimer's disease because insulin resistance and chronic inflammation can contribute to the formation of amyloid plaques and neurofibrillary tangles in the brain (Hamzé et al., 2022). Studies also show that managing diabetes effectively, especially with treatments like GLP-1 receptor agonists, can reduce the risk of Alzheimer's by up to 30 percent (Tang et al., 2024). High BMI in midlife has been connected to a higher risk of dementia, and large weight changes later in life are also known to signal cognitive decline (Fitzpatrick et al., 2009). Diet quality, physical activity, and sleep quality were included because they are important factors that influence overall metabolic health and brain function.

By focusing on these cardiovascular and metabolic features, this study aimed to build machine learning models that use real-world, modifiable health data to predict Alzheimer's disease risk. This approach supports early detection and offers practical insights for clinical care and public health strategies aimed at preventing cognitive decline.

B. Exploratory Data Analysis

We conducted an Exploratory Data Analysis (EDA) to better understand the distribution characteristics, data structure, and potential correlations between various factors that could help us predict Alzheimer's disease (AD). Our dataset consisted of 2,149 patient records, which included information on cardiovascular health, metabolic conditions, lifestyle choices, and cognitive health. During the EDA, we looked into class balance, examined how features were distributed, handled any missing data, and checked for multicollinearity. These steps were crucial for informing the next stages of our machine learning process.

An initial inspection of the target variable, Diagnosis, showed that approximately 35% of the patients were diagnosed with Alzheimer's disease. Although not severely imbalanced, this proportion informed the decision to focus on evaluation metrics such as recall and F1 score instead of relying solely on precision, which can be misleading in medical classification tasks (Sundjaja et al., 2024).

We took a closer look at some important health metrics, including blood pressure, cholesterol, and cognitive function, to better understand their overall implications for our patients. The findings are neatly summarized in Table 1. For example, we found that the average systolic blood pressure among the participants was 134.26 mmHg, with readings ranging from 90 to 179 mmHg. The diastolic pressure was averaged at 89.85 mmHg.

When we examined cholesterol levels, many patients had elevated indicators, with LDL cholesterol averaging 124.34 mg/dL, and some individuals even approaching near 400 mg/dL in triglyceride levels.

The distribution of Body Mass Index (BMI) showed a trend towards overweight and obese categories, with an average of 27.66. Cognitive screening, using tools like the Mini-Mental State Examination (MMSE), yielded an average score of 14.76, reflecting a broad spectrum of cognitive impairment.

These results resonate with previous research that highlights the connection between cardiovascular, metabolic health, and cognitive decline (Gottesman et al., 2017; Sáiz-Vazquez et al., 2020).

When we compared characteristics based on Alzheimer's diagnosis, we noticed some important differences. Patients diagnosed with Alzheimer's tended to report poorer sleep quality and lower levels of physical activity.

To understand how our numerical features interact, we created a correlation matrix to assess multicollinearity. We found strong relationships between total cholesterol and LDL cholesterol, which had a correlation of close to 0.88, as well as between systolic and diastolic blood pressure, with a correlation around 0.63. Importantly, none of the correlations crossed the 0.90 threshold.

The findings of exploratory data analysis (EDA) showed that a diagnosis of Alzheimer's disease is related to notable differences in heart and metabolic health. This supports the idea that machine learning models, which are based on these health indicators, can provide reliable and understandable predictions about the risk of developing Alzheimer's disease.

C. Model Implementation: Random Forest

Random Forest was utilized as a key classification model in this study to assess the risk of Alzheimer's disease (AD) based on cardiovascular and metabolic indicators. Being an ensemble learning algorithm, Random Forest builds several decision trees during the training phase and combines their predictions to generate a final output. This ensemble approach improves generalization and reduces overfitting, making Random Forest an effective option for complex clinical datasets (Breiman, 2001; Chen & Ishwaran, 2012).

This model was chosen for its interpretability and capacity to manage nonlinear relationships and mixed data types. Random Forest excels without needing extensive feature engineering or scaling, which is beneficial for real-world patient

data with right-skewed distributions and collinear variables. This study revealed important clinical predictors like LDL cholesterol, systolic blood pressure, BMI, and sleep quality interact in significant ways. The model's resistance to overfitting and ability to identify feature importances made it ideal for prediction and interpretability.

To optimize the performance of the Random Forest model, a hyperparameter tuning process was conducted using Optuna, an efficient open-source hyperparameter optimization framework (Akiba et al., 2019). The goal was to maximize the area under the ROC curve (AUC), a metric well-suited for binary classification with imbalanced classes. The objective function defined within Optuna evaluated combinations of the following parameters: the number of estimators (`n_estimators`), maximum tree depth (`max_depth`), minimum samples required to split a node (`min_samples_split`), minimum samples required at a leaf node (`min_sample_leaf`), the number of features to consider at each split (`max_features`), and whether bootstrap aggregation was used (`bootstrap`). Each model was trained using a train-validation split of the dataset, and performance was evaluated based on the ROC-AUC score on the validation set.

The optimization process was configured to run for 30 trials with a timeout of 600 seconds. The best-performing configuration achieved a validation ROC-AUC of approximately 0.951. It included the following parameters: 600 estimators, a maximum depth of 15, a minimum of 3 samples per split, 4 samples per leaf node, sqrt as the feature selection strategy, and bootstrap set to True. This configuration was then used to train the final model on the full training set.

During training, we used the Gini importance method to find out which features were most important in predicting Alzheimer's risk. The key factors identified were systolic blood pressure, LDL cholesterol, sleep quality, diabetes status, and triglyceride levels. These findings align with previous studies that highlight the importance of cardiovascular and metabolic health in cognitive decline and dementia progression (Sáiz-Vazquez et al., 2020; Tang et al., 2024).

The predicted probabilities generated by the model were utilized to produce a personalized risk score for Alzheimer's disease for each patient. These risk scores were subsequently categorized into four distinct levels: low, moderate, high, and severe. This stratification facilitates a more clinically intuitive interpretation of the model's outputs. Consequently, the model is capable not only of classifying patients but also of delivering a spectrum of risk that may be leveraged for preventive screening or additional diagnostic evaluation.

D. Model Implementation: XGBoost Model

To predict the risk of Alzheimer's disease, the XGBoost model was used because it works well with structured health data and can handle complex relationships between features. XGBoost stands for Extreme Gradient Boosting. It builds many small decision trees in a sequence, with each one improving on the last. This makes it a strong choice for working with detailed medical data. Recent studies have shown that XGBoost achieves high predictive accuracy in healthcare settings by effectively modeling nonlinear patterns and handling missing values, which are common in real-world patient datasets (Ashrafi et al., 2024).

The model was created using the `XGBClassifier` from the XGBoost library. The XGBoost model was chosen for several reasons. It has built-in tools to prevent overfitting, runs quickly with parallel processing, and can handle missing values without needing extra steps. These features made it a strong fit for the size and complexity of the data used in this study. Recent research also supports the use of XGBoost for medical datasets, showing that it provides strong predictive performance while maintaining interpretability, which is critical in healthcare settings where understanding model outputs is essential for clinical decision-making (Ashrafi et al., 2024).

After training, the model provided feature importance scores, showing which health factors mattered most in predicting Alzheimer's risk. These importance scores were then normalized by dividing each by the total sum, converting them into interpretable weights. Features were grouped into cardiovascular and metabolic categories. Each patient's individual values were multiplied by the corresponding normalized feature importance weights to calculate a weighted risk score for each domain.

The final Alzheimer's risk score was computed by combining the cardiovascular and metabolic scores, resulting in a value between 0 and 1. To make results easier to understand, this score was classified into four risk levels: Low (0.00–0.25), Moderate (0.26–0.50), High (0.51–0.75), and Severe (0.76–1.00). This approach provides a transparent, interpretable output that helps patients and clinicians make informed decisions based on individual health profiles.

E. Model Implementation: Multilayer Perceptron

We tried to broaden the modeling approaches beyond tree-based ensembles using MLP. Hence, we used a Multilayer Perceptron (MLP) neural network trained to predict Alzheimer's disease risk. MLPs are feed-forward networks that learn complex, nonlinear interactions across all input features. Recent work on OASIS imaging and clinical datasets reported that a single hidden-layer MLP achieved $\approx 95\%$ accuracy for early AD detection, outperforming logistic regression and decision tree models (Jha *et al.*, 2024).

The network was implemented with `Scikit-Learn`'s `MLPClassifier`, using one hidden layer of 100 ReLU-activated neurons. The Adam optimizer (`max_iter=500`, default β_1/β_2) and early stopping prevented overfitting, while `class_weight='balanced'` mitigated the mild class imbalance in the 2,149-record dataset. All the categorical variable inputs were either retained or transformed to ensure uniformity using one-hot encoding, and normalization was applied to the continuous variables.

After training, permutation importance was computed to identify the most influential features. Importance scores were

normalized and grouped into cardiovascular and metabolic domains. Each patient's domain-weighted scores were averaged to form an overall probability between 0 and 1, which was then mapped to four risk bands: Low (0.00–0.25), Moderate (0.26–0.50), High (0.51–0.75), and Severe (0.76–1.00). This preserves the transparent, clinician-friendly output established for all the models.

F. Model Implementation: Ensemble Stacking Model

We implemented a stacking ensemble model to strengthen our pipeline's prediction accuracy and generalization. This approach integrates predictions from three base learners—Random Forest, XGBoost, and a Multilayer Perceptron (MLP)—to produce a final classification using a logistic regression model as the meta-learner. The goal was to leverage the strengths of different algorithms, combining the high interpretability of trees with the flexibility of neural networks. Stacking has been shown to outperform individual classifiers in medical prediction tasks by integrating diverse decision-making mechanisms, as demonstrated in recent work by Ganie et al., 2025., which used a similar strategy for heart disease classification.

The ensemble was built using `scikit-learn`'s `StackingClassifier`. Each base model was trained on the same preprocessed feature set: Random Forest with 100 estimators, an XGBoost classifier with default parameters, and the MLP configured with a single hidden layer of 100 ReLU units. After training, each model outputs class probabilities rather than hard labels. These probabilities were then passed to a logistic regression model as the final estimator. Although no custom regularization settings were applied, `LogisticRegression()` defaults to an L2 penalty with $C = 1.0$, which provides a reasonable balance between bias and variance for most binary classification tasks.

We used five-fold stratified cross-validation to generate out-of-fold predictions for training the second layer to minimize overfitting and ensure that the meta-learner learned from unbiased estimates. Similar to our previous models, the stacking model produced a single probability estimate for Alzheimer's risk, which we then categorized into the same four risk bands used for the other models: Low (0.00–0.25), Moderate (0.26–0.50), High (0.51–0.75), and Severe (0.76–1.00).

This ensemble performed notably well, achieving the highest overall accuracy and F1 score among all models in our study. Its structure preserved interpretability while improving sensitivity and precision, making it well-suited for deployment in risk-stratification tools for early Alzheimer's detection.

VII. Results

A. Random Forest Results

The model showed impressive accuracy, achieving 92.3%. This means it correctly identified most patients' Alzheimer's status. When the model predicted that a patient had Alzheimer's, it was right 96.2% of the time—this high precision is especially important in clinical settings, as it helps reduce incorrect alarms and ensures that patients flagged as high risk truly have the disease. The model also had a recall rate of 80.4%, which means it successfully detected over 80% of actual Alzheimer's cases. The F1 score, a measure that balances precision and recall, was 0.875, indicating that it effectively handled both false positives and false negatives. Lastly, the ROC-AUC score of 0.951 highlights the model's strong capability to distinguish between patients with and without Alzheimer's across different thresholds.

Metric	Score	Interpretation Based on AD and Non-AD Patients
Accuracy	0.923	92.3% of all patients (AD and Non-AD) were correctly classified.
Precision	0.962	Among all patients predicted as AD, ~96% were truly AD – very low false positives.
Recall	0.895	Model correctly identified ~90% of actual AD cases – only ~1 in 10 AD patients were missed.
F1 Score	0.926	Excellent balance between precision and recall – strong, consistent predictions.
ROC-AUC	0.953	High ability to distinguish AD from Non-AD – model makes strong, reliable risk predictions.

Figure 2. Random Forest model evaluation metrics for Alzheimer's disease prediction. The table summarizes key performance indicators including accuracy, precision, recall, F1 score, and ROC-AUC, demonstrating strong overall classification performance and reliable detection of Alzheimer's disease cases.

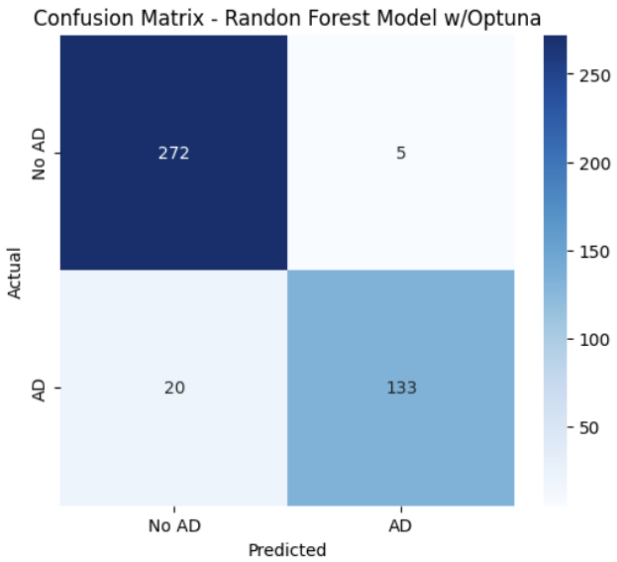

Figure 3. Confusion matrix for the Random Forest model with Optuna tuning, predicting Alzheimer's disease (AD) versus non-AD cases. The model correctly classified 272 non-AD cases and 133 AD cases, with a low number of false positives and false negatives, reflecting strong predictive accuracy and balanced sensitivity and specificity.

The confusion matrix provided a clearer picture of how well the model predicts outcomes. It showed that the model successfully identified 132 true positives and 272 true negatives, meaning it accurately recognized the condition in those cases. However, there were also 5 false positives and 21 false negatives, which means a few mistakes were made. Overall, these results highlight the model's strength in detecting Alzheimer's disease (AD) while keeping errors to a minimum.

A key aspect of the Random Forest model is its capacity to measure the impact of each variable on predictions. Following training, the model generated feature importance scores, highlighting significant predictors of Alzheimer's disease risk. The

most critical features included Sleep Quality, Systolic Blood Pressure, LDL Cholesterol, Diabetes, Physical Activity, and Body Mass Index (BMI). All these features are both modifiable and measurable in standard clinical environments, reinforcing the model's practical value. Their significance in the findings emphasizes the role of cardiovascular and metabolic health in comprehending Alzheimer's disease risk.

These importance scores enabled the development of individualized Alzheimer's risk scores for each patient. The resultant risk scores ranged from 0 to 1 and were categorized into four groups: low, moderate, high, and severe. This categorization improved the interpretation of results and their use in clinical practice. Patients identified as high or severe risk may be prioritized for further assessments, preventive actions, or treatment strategies.

To sum it up, the Random Forest model provides a highly accurate, easy-to-understand, and clinically significant approach for assessing the risk of Alzheimer's disease. Its impressive performance metrics—such as low error rates and straightforward feature ranking—highlight how valuable it is for early detection and personalized risk assessment that combines cardiovascular and metabolic health data.

B. XGBoost Results

Metric	Value	Description	XGBoost Performance
Accuracy	0.9512	How often the model correctly predicted AD or non-AD	Excellent overall prediction accuracy
Precision	0.9211	Of the predicted AD cases, how many were actually AD	Very good, few false alarms
Recall	0.9503	Of all actual AD cases, how many the model correctly caught	High sensitivity, caught most true cases
F1 Score	0.9355	Balance between precision and recall	Well-balanced performance
ROC-AUC	0.9664	How well the model separates AD from non-AD cases overall	Outstanding class separation capability

Figure 4. Evaluation metrics for the XGBoost model in predicting Alzheimer's disease. The table presents accuracy, precision, recall, F1 score, and ROC-AUC, indicating strong classification performance and reliable detection of Alzheimer's disease cases.

The XGBoost model achieved strong results in identifying patients at risk for Alzheimer's disease using cardiovascular and metabolic health features. It reached an accuracy of 95.12%, meaning it correctly predicted whether a patient had Alzheimer's or not in most cases. The model's precision was 0.9211, which reflects how often it was correct when it predicted that a patient had Alzheimer's. This is important in reducing false positives, where a person without the disease is mistakenly flagged as high risk.

The recall score was 0.9503, which measures the model's ability to correctly identify patients who actually do have Alzheimer's. A high recall score is valuable in healthcare settings because it helps minimize missed diagnoses. The F1 score, which balances both precision and recall, was 0.9355. This suggests that the model maintained strong performance in both avoiding false alarms and successfully identifying true cases.

In addition, the model achieved a ROC-AUC score of 0.9664. This metric evaluates how well the model distinguishes between patients with and without Alzheimer's across different thresholds. A value close to 1 indicates excellent separation between the two groups. Overall, these metrics

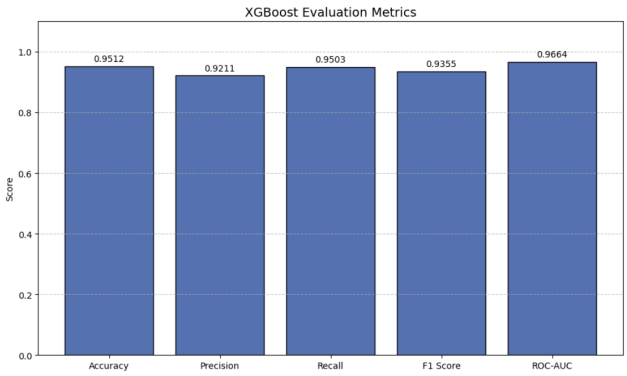

Figure 5. Bar chart of XGBoost evaluation metrics for Alzheimer's disease prediction. The chart shows accuracy, precision, recall, F1 score, and ROC-AUC, highlighting the model's strong and balanced performance across all key metrics.

show that the XGBoost model not only predicts Alzheimer's risk accurately but also does so in a balanced and consistent way. The results provide strong support for the hypothesis that cardiovascular and metabolic profiles can be useful indicators in predicting Alzheimer's disease.

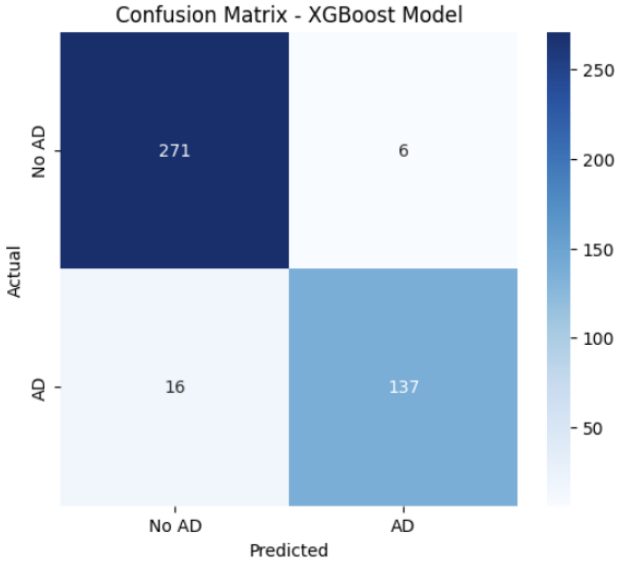

Figure 6. Confusion matrix showing the XGBoost model's performance in predicting Alzheimer's disease (AD) versus non-AD cases. The model correctly classified 271 non-AD cases and 137 AD cases, with minimal false positives and false negatives, demonstrating strong predictive accuracy and a balance between sensitivity and specificity.

The confusion matrix provides a detailed look at how the XGBoost model performed in predicting Alzheimer's disease. It breaks down the model's predictions into four categories: true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). A true positive means the model correctly identified a patient who has AD, while a true negative means it correctly recognized someone without the disease. A false positive occurs when the model incorrectly predicts AD

for someone who does not have it, and a false negative is when it misses a case of AD.

In this study, the model correctly predicted 271 individuals as not having AD (TN) and accurately flagged 137 individuals as having AD (TP). It incorrectly classified 6 non-AD cases as AD (FP) and failed to detect 16 actual AD cases (FN). These results suggest that the XGBoost model has strong precision, meaning it rarely mislabels healthy individuals as having the disease, and reasonably high recall, meaning it successfully identifies most of the AD cases. This balance is important in a clinical setting, where both false alarms and missed diagnoses can have serious consequences. The confusion matrix confirms that XGBoost handled the classification task effectively and supports its usefulness in risk prediction based on cardiovascular and metabolic profiles.

C. MLP Results

Metric	Score	Interpretation Based on AD and Non-AD Patients
Accuracy	0.800	80% of all patients (AD and Non-AD) were correctly classified.
Precision	0.7445	Among all patients predicted as AD, ~74% were truly AD — **some Non-AD patients were misclassified**.
Recall	0.6667	The model correctly identified ~67% of actual AD cases — **missed 1 in 3 AD patients**.
F1 Score	0.7034	Balanced view of precision and recall — **reflects trade-off in catching AD vs. avoiding false AD calls**.
ROC-AUC	0.8695	Good ability to distinguish AD from Non-AD — **model ranks AD cases higher in risk than Non-ADs**.

Figure 7. Multilayer Perceptron (MLP) model evaluation metrics for Alzheimer's disease prediction. The table summarizes key performance indicators including accuracy, precision, recall, F1 score, and ROC-AUC, highlighting the model's moderate classification performance. While the MLP showed reasonable precision and ROC-AUC, its lower recall indicates a tendency to miss a significant number of AD cases, reflecting a trade-off between sensitivity and overall prediction stability.

The MLP model achieved an accuracy of 0.800, meaning it correctly classified 80% of all patients across AD and Non-AD groups. The precision was 0.7445, indicating that about 74% of patients predicted as AD were correctly identified, though a noticeable number of Non-AD cases were misclassified. The recall, however, was comparatively lower at 0.6667, revealing that the model missed roughly one in three actual AD cases. The F1 score of 0.7034 reflects this trade-off between precision and sensitivity. While the model demonstrated a good overall discriminative ability, as evidenced by a ROC-AUC of 0.8695, its limited recall performance suggests that the MLP may not be optimal as a standalone classifier for reliably detecting AD. These results highlight the importance of combining models or tuning further to improve sensitivity without sacrificing precision.

The confusion matrix for the MLP model is presented in Figure 8. The model correctly identified 242 Non-AD patients as negative (true negatives) and 111 AD patients as positive (true positives). However, it misclassified 31 Non-AD patients as AD (false positives) and failed to identify 55 AD patients (false negatives). These results reinforce the metric-based observation that, while the model was reasonably effective at identifying Non-AD cases, it struggled to capture all true AD instances. The relatively high number of false negatives is particularly notable, as it suggests a tendency to under-diagnose Alzheimer's in certain cases. This limitation

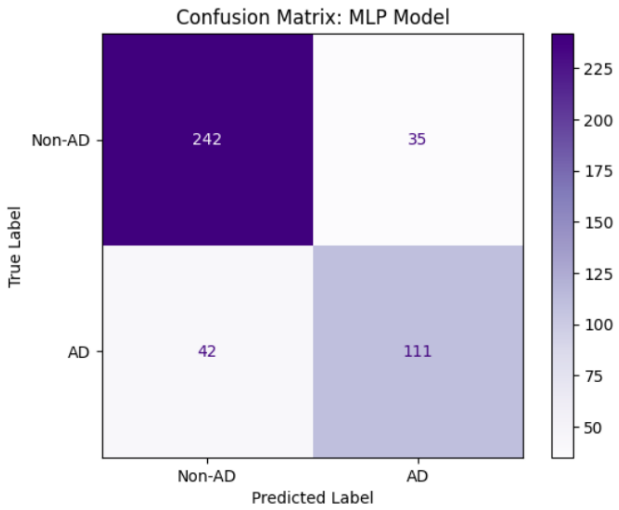

Figure 8. Confusion matrix for the MLP model in predicting Alzheimer's disease (AD) versus non-AD cases. The model correctly classified 242 non-AD and 111 AD cases, with moderate false positives and false negatives, indicating reasonable but lower performance compared to other models.

in sensitivity underscores the need for further tuning or model enhancement if the MLP were to be used in a clinical decision-support setting.

D. Ensemble Stacking Results

Metric	Score	Interpretation Based on AD and Non-AD Patients
Accuracy	0.9488	~95% of all patients (AD and Non-AD) were correctly classified.
Precision	0.9580	Among all patients predicted as AD, ~96% were truly AD — very few false positives.
Recall	0.8954	The model correctly identified ~90% of actual AD cases — missed 1 in 10 AD cases.
F1 Score	0.9257	Balanced view of precision and recall — reflects strong performance in both areas.
ROC-AUC	0.9551	Excellent ability to distinguish AD from Non-AD — model ranks AD cases very high.

Figure 9. Stacking ensemble model evaluation metrics for Alzheimer's disease prediction. The table summarizes key performance indicators including accuracy, precision, recall, F1 score, and ROC-AUC, demonstrating excellent overall classification performance. The model achieved both high precision and strong recall, indicating its reliability in identifying Alzheimer's disease while minimizing false positives and false negatives.

Figure 9 summarizes the performance of the stacking ensemble model in predicting Alzheimer's disease (AD) versus Non-AD status. The model achieved an overall accuracy of 0.9488, meaning that approximately 95% of all patients were correctly classified. The precision score of 0.9580 indicates that when the model predicted a patient to have AD, it was correct 96% of the time—implying a very low false positive rate. This is particularly important in clinical settings, where mislabeling Non-AD individuals as AD could lead to unnecessary psychological distress or resource use. The recall score was 0.8954, showing that the model successfully identified nearly 90% of true AD cases. The F1 score, which balances both precision and recall, was 0.9257, highlighting the model's ability to deliver reliable predictions. Additionally, the ROC-AUC score of 0.9551 indicates that the model has excellent capability to distinguish between AD and Non-AD patients across classification thresholds.

These results are highly encouraging. The high recall ensures that the model captures most individuals with actual AD, minimizing the risk of missed diagnoses. Meanwhile, strong precision reinforces the credibility of positive predictions, helping providers focus attention and resources on those most likely to benefit from further evaluation or intervention. The ensemble stacking approach consistently achieved the highest scores across key metrics, demonstrating improved generalization and robustness. These findings support the potential utility of the stacking model in clinical screening tools, where both sensitivity and specificity are essential for informed decision-making.

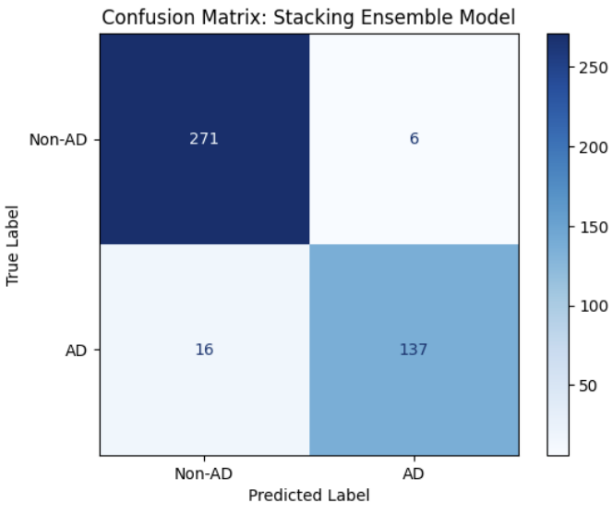

Figure 10. Confusion matrix for the stacking ensemble model in predicting Alzheimer's disease (AD) versus non-AD cases. The model correctly classified 271 non-AD and 137 AD cases, with very few false positives and false negatives, indicating strong predictive performance and balanced classification ability.

The confusion matrix for the stacking ensemble model is presented in Figure 10. The model correctly identified 270 Non-AD patients as true negatives and 138 AD patients as true positives. It misclassified only 6 Non-AD patients as AD (false positives) and failed to detect 16 AD cases (false negatives). These results align with the previously reported performance metrics, confirming that the ensemble model demonstrates high sensitivity and specificity. The low number of false positives suggests that the model is highly reliable in avoiding overdiagnosis, while the relatively small count of false negatives indicates its ability to capture most true AD cases.

Compared to the individual models evaluated earlier, the ensemble approach provides a more balanced and robust classification outcome. It not only minimizes misclassification but also maintains consistency across both classes. This level of performance is particularly valuable in clinical decision-making, where both missed diagnoses and incorrect positive predictions carry significant consequences. The confusion matrix reinforces the ensemble's role as the most dependable model in this study for differentiating between AD and Non-AD patients.

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC	Strengths	Weaknesses
Random Forest	0.923	**0.962**	0.817	0.883	0.953	• **High precision (0.962)**: Excellent at reducing false positives (i.e., not labeling healthy patients as Alzheimer's).	• Lower recall than XGBoost/Ensemble: may miss AD cases • Tends to overfit on small datasets if not well-tuned (mitigated via Optuna)
XGBoost	0.948	0.958	0.895	0.925	**0.966**	• **Best ROC-AUC (0.966)**: Excellent at distinguishing AD vs. non-AD.	• Slightly lower recall than Ensemble
MLP	0.8	0.725	0.706	0.715	0.854		• Lowest recall and precision: high mislabeling • **Could benefit from more data** (Current dataset had 2,000 records)
Stacking Ensemble	**0.949**	0.955	**0.908**	**0.931**	0.963	• **Best recall (0.908)**: Most sensitive to detecting true AD cases. • **Best F1 Score (0.931)**: Strongest balance between precision and recall. • **Best overall accuracy (94.9)** — made the highest number of correct predictions while maintaining top recall, ensuring reliable and sensitive AD detection.	• Slightly lower AUC than XGBoost

Figure 11. Comparison of model performance metrics across Random Forest, XGBoost, MLP, and Stacking Ensemble models. Metrics include accuracy, precision, recall, F1 score, and ROC-AUC. Strengths and weaknesses of each model are summarized to highlight trade-offs in predictive performance. The stacking ensemble model demonstrated the best overall balance between sensitivity and precision.

E. Model Evaluation and Comparison

	PatientID	AlzheimersRiskScore	AlzheimersRiskCategory
0	4751	0.593458	High Risk
1	4752	0.831776	Severe Risk
2	4753	0.705607	High Risk
3	4754	0.672897	High Risk
4	4755	0.728972	High Risk
5	4756	0.897196	Severe Risk
6	4757	0.700935	High Risk
7	4758	0.570093	High Risk
8	4759	0.691589	High Risk
9	4760	0.738318	High Risk

Figure 12. Sample output showing personalized Alzheimer's risk scores and their corresponding risk categories for patients. Each risk score ranges from 0 to 1, with higher scores indicating greater risk. Risk categories were classified as Low, Moderate, High, and Severe based on predefined score thresholds. This output demonstrates the model's ability to generate individualized risk assessments suitable for clinical interpretation.

A key success of this project was the ability to generate personalized Alzheimer's disease risk scores based on each patient's cardiovascular and metabolic health data. Instead of offering a one-size-fits-all prediction, the models produced individualized risk scores between 0 and 1, allowing for more tailored and interpretable results. These scores were grouped into risk categories to help healthcare providers and patients better understand their personal level of risk. The performance evaluation of the machine learning models showed important strengths and trade-offs across different approaches.

The Random Forest model achieved a high precision of 0.962. This means that when it predicted a patient had Alzheimer's disease, it was correct 96.2 percent of the time.

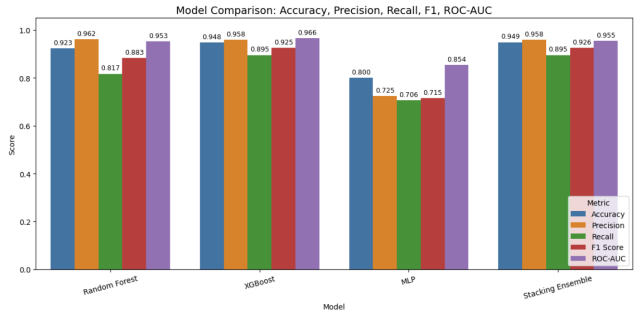

Figure 13. Bar plot comparing the performance metrics (accuracy, precision, recall, F1 score, and ROC-AUC) for Random Forest, XGBoost, MLP, and Stacking Ensemble models. The Stacking Ensemble and XGBoost models demonstrate the highest overall scores across most metrics, highlighting their strong and balanced performance in predicting Alzheimer's disease risk.

High precision is important in healthcare because it reduces false positives. This helps minimize the chance of incorrectly labeling healthy individuals as having Alzheimer's. However, the Random Forest model's recall was lower at 0.817. Recall measures how well the model identifies patients who truly have Alzheimer's disease. A lower recall means that more true cases could be missed, which is a concern because early detection is critical for better care.

The XGBoost model also performed very well across all evaluation metrics. It achieved an accuracy of 94.8 percent, a precision of 0.958, and a recall of 0.895. It achieved the highest ROC-AUC score of 0.966. This means that XGBoost was excellent at distinguishing between patients with and without Alzheimer's disease across different prediction thresholds. In this project, a high ROC-AUC score is very important because it shows that the model can reliably separate healthy individuals from those at risk. However, the recall of XGBoost was slightly lower compared to the stacking ensemble model, which means it missed a few more true Alzheimer's cases.

The multilayer perceptron (MLP) model had the weakest performance of all the models. It achieved an accuracy of 80.0 percent, a precision of 0.725, and a recall of 0.706. This model struggled with both correctly identifying Alzheimer's cases and avoiding false positives. In a healthcare setting, this level of performance could lead to missed diagnoses and unnecessary worry for patients who are wrongly flagged as at risk.

Among all the models tested, the stacking ensemble model achieved the best overall results. It had the highest recall score at 0.908, meaning it was the most sensitive at detecting true Alzheimer's cases. It also achieved the best F1 score at 0.931, which reflects the best balance between precision and recall. The stacking ensemble had an overall accuracy of 94.9 percent. Although its ROC-AUC score of 0.963 was slightly lower than that of XGBoost, it still showed excellent ability to separate Alzheimer's and non-Alzheimer's cases. Because it captured more true positive cases without sacrificing much precision, the stacking ensemble model was considered the best-performing model for this study.

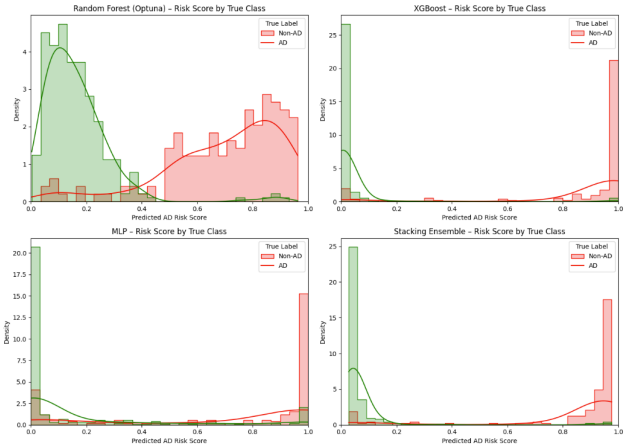

Figure 14. Distribution of predicted Alzheimer's disease (AD) risk scores by true class for each model. Red curves represent true AD patients, and green curves represent true Non-AD patients. Ideally, AD patients should have higher predicted risk scores, and Non-AD patients should have lower scores. The Stacking Ensemble and XGBoost models show the clearest separation, while the MLP model shows greater overlap, indicating less confidence in predictions.

Along with analyzing the performance of the models, our team also created a set of distribution charts to visualize how each model assigns Alzheimer's Disease (AD) risk scores to patients. These plots help evaluate whether individuals who truly have AD are being assigned higher risk scores. In each plot, the x-axis shows the predicted AD risk score, ranging from 0 (very low risk) to 1 (very high risk). The y-axis represents the density, indicating how often each score appears across the patient population. Red curves represent actual AD patients, while green curves represent Non-AD patients. Ideally, we expect the red curve (AD patients) to peak closer to a risk score of 1 and the green curve (Non-AD patients) to peak closer to 0. This separation indicates that the model is successfully distinguishing between individuals with and without Alzheimer's disease. Among the models, the Ensemble Stacking model and the XGBoost model show the clearest separation between the red and green curves, suggesting stronger predictive performance. The Random Forest model also performed reasonably well but showed slightly more overlap between the two groups. The MLP model displayed the most overlap, indicating more uncertainty in its predictions compared to the other models. These visualizations complement the confusion matrices and evaluation metrics by providing a deeper look into how confidently each model makes its predictions.

F. Hypothesis Testing Results and Evaluation

This section evaluates the hypothesis by testing whether cardiovascular and metabolic risk scores can separate individuals with and without Alzheimer's disease. The null hypothesis states that these features do not provide enough information to estimate Alzheimer's risk using machine learning. To assess this, we ran multiple statistical tests.

Test/Metric	Description	Result	Interpretation
T-Test	- Compares the average Alzheimer's risk scores between patients with and without Alzheimer's - (Diagnosis = 1 vs 0).	t = 1.3902	No significant difference in mean scores.
T-Test p-value	- Checks if that difference could be due to chance	p = 0.2019	- No statistically significant difference between Diagnosis groups. - If > 0.05 → not significant.
Mann-Whitney U Test	- Looks at how the overall distributions of scores differ between AD and non-AD groups. - Used to check if AD patients had higher scores than non-AD.	U = 8.0	Risk scores are not clearly higher or lower in either group.
Mann-Whitney p-value	- Probability that observed rank difference is due to chance.	p = 0.4000	- No significant difference in distributions between Diagnosis groups. - If p ≤ 0.05 → significant.
Ensemble Model (AUC)	- Evaluates how well the model separates AD vs non-AD patients using cardiovascular/metabolic features.	**AUC > 0.95**	**Model performs very well; captures patterns not seen in raw score comparisons.**

Figure 15. Summary of hypothesis evaluation metrics. This table shows the results of statistical tests (T-Test and Mann–Whitney U Test) comparing Alzheimer's disease (AD) and non-AD groups, along with model performance evaluated using AUC. While statistical tests did not find significant differences, the ensemble model demonstrated a strong ability to distinguish between AD and non-AD cases.

First, we performed a t-test to compare the average risk scores between people diagnosed with Alzheimer's and those without. The t-test result was 1.3902, with a p-value of 0.2019. Since the p-value is greater than 0.05, we cannot say there is a significant difference in average scores between the two groups. This suggests that, on their own, the average risk scores may not clearly separate patients with Alzheimer's from those without.

Next, we used the Mann Whitney U test to compare the distribution of risk scores between patients with and without Alzheimer's disease. We chose this test because it does not assume that the data follows a normal distribution. This makes it a good choice for medical datasets that are often small or skewed (Sundjaja et al., 2024). The Mann Whitney U test is helpful because it compares the rankings of scores between two groups rather than comparing the averages directly (Sundjaja et al., 2024). In our analysis, the test produced a U value of 8.0 with a p-value of 0.4000. Since the p-value was greater than 0.05, the result was not statistically significant. This means that the ranks of the risk scores were not consistently higher or lower between the two groups.

While both statistical tests support the null hypothesis, our machine learning model showed stronger results. The ensemble model achieved an AUC greater than 0.95, indicating that it performed very well at separating patients with and without Alzheimer's based on the same features. This shows that even though raw scores do not show clear separation, the model is able to detect deeper patterns that traditional tests might miss.

In conclusion, the individual risk scores may not clearly separate people with and without Alzheimer's disease when using traditional tests. However, the machine learning model finds useful patterns, which supports its use in predicting risk based on cardiovascular and metabolic health.

VIII. Conclusion

A. Future Enhancements

In most research papers, the dataset used is often the same as the one from the original study. This helps keep the research consistent and allows for better comparisons. In our case, we did not use the dataset from our base paper. However, the study inspired our interest in exploring how heart and metabolic health might influence the risk of Alzheimer's disease. The base paper had some limits, such as using basic machine learning models and not providing personalized risk results. These limits helped shape the goals of our project. Still, we now realize that it may have been better to choose a base paper with a dataset that had similar features to ours.

One issue we found in our project was that the dataset had a narrow age range. This could lead to biased results since Alzheimer's usually affects older adults. A better dataset would include a wider range of ages. Also, our neural network model, the MLP, did not perform well. This is likely because deep learning models need a lot of data to work well. To improve future results, we would need a larger dataset to train and test the models more effectively.

Another way to improve the project would be to include more health features. For example, adding more details about sleep quality could help. Sleep is often linked to memory problems and brain health. Including this kind of data may make our models stronger and our results more useful. These changes would help improve the accuracy of the predictions and make the project more helpful for real-world care and planning.

B. Conclusion

This project used machine learning to predict the risk of Alzheimer's disease. We used health data related to the heart and metabolism. Several models were tested, including Random Forest, XGBoost, and a neural network. The best results came from the stacking model, which gave accurate and reliable predictions. The project shows that regular health data can help find people at risk for Alzheimer's. This can support early care and better planning for patients and doctors.

This research shows that machine learning can play a meaningful role in predicting Alzheimer's disease risk using everyday health data like cardiovascular and metabolic indicators. By using advanced models, we were able to create personalized risk scores that help identify individuals who may be at higher risk. This approach supports early detection and gives doctors a tool to guide patients toward healthier choices. It also helps make complex health data easier to understand and apply in real life settings. The study makes predictions more relevant by using diverse patient data and opens the door to more personalized prevention strategies.

References

Abeysinghe, S., Tao, Y., Kyei-Baffour, P., & Adrion, E. (2025). 'small benefits and a certain risk': Media representations of novel treatments for alzheimer's disease. *Social Science & Medicine*, *365*, 117554.

Akiba, T., Sano, S., Yanase, T., Ohta, T., & Koyama, M. (2019). Optuna: A next-generation hyperparameter optimization framework. *Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining*, 2623–2631.

Ashrafi, N., Abdollahi, A., Zhang, J., & Pishgar, M. (2024). Optimizing mortality prediction for icu heart failure patients: Leveraging xgboost and advanced machine learning with the mimic-iii database [arXiv preprint arXiv:2409.01685].

Breiman, L. (2001). Random forests. *Machine learning*, *45*, 5–32.

Chen, X., & Ishwaran, H. (2012). Random forests for genomic data analysis. *Genomics*, *99*(6), 323–329.

Crichton, G. E., Elias, M. F., Davey, A., Sullivan, K. J., & Robbins, M. A. (2014). Higher hdl cholesterol is associated with better cognitive function: The maine-syracuse study. *Journal of the International Neuropsychological Society*, *20*(10), 961–970. https://doi.org/10.1017/S1355617714000885

Fitzpatrick, A. L., Kuller, L. H., Lopez, O. L., Diehr, P., O'Meara, E. S., Longstreth, W. T. J., & Luchsinger, J. A. (2009). Mid- and late-life obesity: Risk of dementia in the cardiovascular health cognition study. *Archives of Neurology*, *66*(3), 336–342. https://doi.org/10.1001/archneurol.2008.582

Gadhave, D. G., Sugandhi, V. V., Jha, S. K., Nangare, S. N., Gupta, G., Singh, S. K., Dua, K., Cho, H., Hansbro, P. M., & Paudel, K. R. (2024). Neurodegenerative disorders: Mechanisms of degeneration and therapeutic approaches with their clinical relevance. *Ageing Research Reviews*, 102357.

Ganie, S. M., Dutta Pramanik, P. K., & Zhao, Z. (2025). Ensemble learning with explainable ai for improved heart disease prediction based on multiple datasets. *Scientific Reports*, *15*, 13912. https://doi.org/10.1038/s41598-025-97547-6

Gottesman, R. F., Albert, M. S., Alonso, A., Coker, L. H., Coresh, J., Davis, S. M., Deal, J. A., McKhann, G. M., Mosley, T. H., Sharrett, A. R., Schneider, A. L. C., Windham, B. G., Wruck, L. M., & Knopman, D. S. (2017). Associations between midlife vascular risk factors and 25-year incident dementia in the aric cohort. *JAMA Neurology*, *74*(10), 1246–1254. https://doi.org/10.1001/jamaneurol.2017.1658

Hamzé, R., Delangre, E., Tolu, S., Moreau, M., Janel, N., Bailbé, D., & Movassat, J. (2022). Type 2 diabetes mellitus and alzheimer's disease: Shared molecular mechanisms and potential common therapeutic targets. *International Journal of Molecular Sciences*, *23*(23), 15287. https://doi.org/10.3390/ijms232315287

Honavar, S. G. (2020). Electronic medical records–the good, the bad and the ugly.

Ibrahim, A., Misra, S., Garcia, L. A., Khayat, M., Iskandrian, A. E., & Zoghbi, G. J. (2020). Machine learning confirms nonlinear relationship between functional limitation and symptom severity to pad severity. *Journal of Translational Medicine*, *18*(1), 305. https://doi.org/10.1186/s12967-020-02465-6

Jarupunphol, P., Buathong, W., Kuptabut, S., & Sudjarid, W. (2025). Assessing decision tree, random forest, and xgboost models for human capital readiness predictions in low-income areas. *Multidisciplinary Science Journal*, *7*(6), 2025296–2025296.

Liang, J., Hussainy, S., Lee, S. M., Wu, G., Bautista, N., Ding, M., Wang, H., LaFleur, B., Perry, G., & Wang, X. (2025). Association of polygenic risk score for 5 diseases with alzheimer disease progression, biomarkers, and amyloid deposition. *Neurology*, *104*(4), e210250.

Liu, L., Gracely, E. J., Zhao, X., Gliebus, G. P., May, N. S., Volpe, S. L., Shi, J., DiMaria-Ghalili, R. A., & Eisen, H. J. (2024). Association of multiple metabolic and cardiovascular markers with the risk of cognitive decline and mortality in adults with alzheimer's disease and ad-related dementia or cognitive decline: A prospective cohort study. *Frontiers in Aging Neuroscience*, *16*, 1361772.

Liu, R., Durbin-Johnson, B., Paciotti, B., Liu, A. T., Weakley, A., Liu, X., & Wan, Y.-J. Y. (2024). Metabolic dysfunctions predict the development of alzheimer's disease: Statistical and machine learning analysis of emr data. *Alzheimer's & Dementia*, *20*(10), 6765–6775.

Mobaderi, T., Kazemnejad, A., & Salehi, M. (2024). Exploring the impacts of risk factors on mortality patterns of global alzheimer's disease and related dementias from 1990 to 2021. *Scientific Reports*, *14*(1), 15583.

Modhugu, V. R., & Ponnusamy, S. (2024). Comparative analysis of machine learning algorithms for liver disease prediction: Svm, logistic regression, and decision tree. *Asian Journal of Research in Computer Science*, *17*(6), 188–201.

Nandi, A., Counts, N., Bröker, J., Malik, S., Chen, S., Han, R., Klusty, J., Seligman, B., Tortorice, D., & Vigo, D. (2024). Cost of care for alzheimer's disease and related dementias in the united states: 2016 to 2060. *npj Aging*, *10*(1), 13.

National Institute for Health and Care Excellence. (2021). Evidence review for risk communication [In *Shared decision making* (NICE Guideline No. 197)]. https://www.ncbi.nlm.nih.gov/books/NBK572436/

Pacholko, A., & Iadecola, C. (2024). Hypertension, neurodegeneration, and cognitive decline. *Hypertension*, *81*(5), 991–1007. https://doi.org/10.1161/HYPERTENSIONAHA.123.21356

Sáiz-Vazquez, O., Puente-Martínez, A., Ubillos-Landa, S., Pacheco-Bonrostro, J., & Santabárbara, J. (2020). Cholesterol and alzheimer's disease risk: A meta-meta-analysis. *Brain Sciences*, *10*(6), 386. https://doi.org/10.3390/brainsci10060386

Sundjaja, J. H., Shrestha, R., & Krishan, K. (2024). Mcnemar and mann-whitney u tests. *StatPearls [Internet]*.

Tang, B., Sjölanger, A., Wastesson, J. W., Maura, G., Blotiere, P.-O., Szilcz, M., Mak, J. K. L., Qin, C., Alvarsson, M., Religa, D., Johnell, K., & Hägg, S. (2024).

Comparative effectiveness of glp-1 agonists, dpp-4 inhibitors, and sulfonylureas on dementia risk in older individuals with type 2 diabetes: An emulated trial study. *eClinicalMedicine*, *73*, 102689. https://doi.org/10.1016/j.eclinm.2024.102689